Renewable and Sustainable Energy Transition 5 (2024) 100079

Contents lists available at ScienceDirect

Renewable and Sustainable Energy Transition

journal homepage: www.journals.elsevier.com/renewable-and-sustainable-energy-transition

Full-length article

Strategizing towards sustainable energy planning: Modeling the mix of future generation technologies for 2050 in Benin

Romain Akpahou [a,*], Flavio Odoi-Yorke [a,c], Lena D. Mensah [a,d], David A. Quansah [a,d], Francis Kemausuor [b,d]

[a] Department of Mechanical Engineering, Kwame Nkrumah University of Science and Technology (KNUST), Ghana
[b] Department of Agricultural and Biosystems Engineering, Kwame Nkrumah University of Science and Technology (KNUST), Ghana
[c] Department of Renewable Energy Technology, Cape Coast Technical University, Cape Coast, Ghana
[d] The Brew-Hammond Energy Centre, Kwame Nkrumah University of Science and Technology (KNUST), Ghana

ARTICLE INFO

Keywords:
Benin Republic
Renewable energy resources
EnergyPLAN
Energy planning
Sustainable energy transition

ABSTRACT

The Benin energy sector faces serious challenges, including an unfavorable energy mix with regular power shortages, erratic power outages, reliance on electricity imports, and dependence on traditional cooking stoves. This study has investigated strategies critical for Benin to employ to achieve 24.6 %, 44 %, and 100 % renewable energy (RE) integration targets in the final electricity mix in 2025, 2030, and 2050, respectively. This study used the EnergyPLAN model to develop different energy scenarios suitable for Benin to achieve its proposed RE penetration target. A combination of natural gas (NG) with solar photovoltaic (PV), wind energy, hydropower, and concentrated solar power (CSP) is used to develop three scenarios for RE integration namely the government targets scenario, 2 % RE per year scenario and 50 % RE in 2050 scenario. The results show that the government targets scenario is too ambitious because of the current trend and pace of developing the energy sector. Moreover, a combination of 563 MW of NG, 125 MW of PV, 200 MW of wind, 600 MW of hydropower, and 60 MW of CSP would achieve 50 % RE by 2050 under the 50 % RE scenario. This scenario would decrease CO_2 emissions by 50 % with no CEEP generation. Furthermore, the total electricity generation from MSW in Benin is estimated to be 0.232, 0.3215, and 1.16 TWh/yr in 2025, 2030, and 2050, respectively. The study's findings could help decision-makers and stakeholders make informed decisions to promote the integration of RE resources in the Benin Republic.

1. Introduction

The United nations (UN) introduced the sustainable development goals (SDGs) to guide social development endeavors. A key focal point of these goals is the imperative to transition towards innovative energy systems centered around renewable energy (RE) resources [1]. Several nations across the globe have actively engaged in formulating strategies to shift their energy frameworks towards greater sustainability to meet future demands [2]. The global adoption of renewable energy technologies not only corresponds with the broader objectives of addressing

List of abbreviations: UN, United Nations; SDG, Sustainable Development Goal; RE, Renewable Energy; IRENA, International Renewable Energy Agency; GHG, Greenhouse gas; ECOWAS, Economic Community of West African States; PV, Photovoltaic; PRODERE, Renewable Energy Development Program; PROVES, Solar Energy Promotion Project; MW, MegaWatt; SA, South Africa; SSA, Sub-Saharan Africa; NDC, National Determined Contribution; GWh, Gigawatt hour; TWh, Terawatt hour; NG, Natural gas; CEEP, Critical Excess Electricity Production; PPI, Power Production Importation; P_W, Maximum power output from wind turbine; P_{HY}, Power output from hydropower plant; P_{PV}, Power output from solar photovoltaic plant; ε, Waste collection efficiency; η, Combined transmission and electrical efficiency; ρ, Water density; C_P, Power coefficient; A, Cross-sectoral area of turbine's rotor; V, Wind speed; H(t), Hourly solar insolation; S, Solar module surface area; η_{PV}, Efficiency of solar module; Q, Water flow rate; H, Fall height; g, Acceleration of gravity; R_f, Renewable factor; $P_{non\text{-}renewable}$, Energy output from non-renewable sources; $P_{renewable}$, Energy output from renewable sources; O&M, Operation and Maintenance; M$, Million United States dollars; B$, Billion United States dollars; MSW, Municipal solid waste; CH4, Methane; L_O, Biogas production factor; L_{CH4}, Lower heating value of methane; E_{CH4}, Gross energy production of methane; f_{LF}, Fraction of municipal solid waste sent to landfills; N, Urban population; q, Quantity of waste collected; E_{el}, Annual electricity generation from methane; CO_2, Carbon dioxide; $MtCO_2$, Million tons of carbon dioxide.

* Corresponding author.
E-mail address: romain.akpahou@aims.ac.rw (R. Akpahou).

https://doi.org/10.1016/j.rset.2024.100079
Received 14 October 2023; Received in revised form 22 December 2023; Accepted 8 January 2024
Available online 11 January 2024

environmental issues and enhancing public health but is also vital in attaining universal energy access and promoting comprehensive sustainable development [3].

However, it is worth mentioning that a substantial portion of the global population still lacks reliable and affordable energy access, thereby relying on fossil fuel-based technologies for their daily energy requirements [4,5]. The primary concern associated with fossil fuels is their detrimental greenhouse gas (GHG) emissions, which are major contributors to climate change and air pollution [6]. Electricity access is a significant challenge in developing nations, notably the sub-saharan African (SSA) region. For instance, it is estimated that about 77 % of individuals without access to electricity reside in SSA [4]. In a recent report by Africa energy review [6], energy demand in Africa is escalating at an annual growth rate of 3 %, the highest among all continents, despite a considerable energy supply deficit. Therefore, developing strategies to seamlessly incorporate locally available renewable energy resources into a nation's overall energy mix has become an urgent priority [7].

Benin Republic is a developing country in the SSA region with a significant energy access situation. More than half of the population lacks access to electricity [8]. Despite the country's abundant energy resources, it still relies on traditional fossil-fuel energy sources [9]. Nigeria and Ghana are nearby nations that supply electricity to Benin's cities [10]. The country's electricity consumption was about 94 kWh per capita in 2020, against 116 kWh on average in the economic community of west African states (ECOWAS) region [9]. The nation's energy mix comprises local electricity generation, with the central Maria Gléta power plant contributing 127 MW, alongside other in-country production sources [11]. Nevertheless, in Benin's energy sector, RE source contribution is poor and depends mainly on natural gas and petroleum products for its production.

Benin is endowed with RE resources. solar energy, wind energy, biomass energy, and hydro are the most dominant sources in the country [12]. Benin government's recent efforts have led to the implementation of numerous RE projects, such as constructing a 25 MW solar power plant, the first in the country. Besides, the government has developed plans to improve energy access and reduce the dependence on electricity imports to accelerate Benin's energy transition [13]. Some of these initiatives include off-grid rural electrification projects, the renewable energy development program (PRODERE), the solar energy promotion project (PROVES) [14], and the development of a national policy for developing RE in Benin by 2035 [15]. In addition, the millennium challenge account (MCA) in Benin is trying to finance and deploy different solar projects in Benin [12]. Nevertheless, electricity generation from biomass has not been developed in the country. Biomass resources are traditionally used for cooking or burning in the open air at storage sites [16].

In the ECOWAS region, all member countries have set goals to integrate RE into their energy supply mix [1]. Given this, ECOWAS aims to integrate at least 48 % of RE by 2030 at the regional level. In the case of Benin, the target is to integrate at least 24.6 % by 2025 [8] and 44 % by 2030 with a 100 % electricity access rate [17]. It is worth mentioning that Benin's final electricity demand is projected to reach 2.7 TWh and 3.74 TWh, respectively, in 2025 and 2030 [1]. Therefore, this study intends to propose energy planning strategies to assist the government in meeting demand and targets by using locally available RE sources. The study aims to propose pathways to meet future electricity demand by using solar PV, wind energy, hydropower, and concentrated solar power technologies. The EnergyPLAN energy model is used to analyze the energy, environmental, and economic impacts of various energy strategies in the Benin Republic. In addition, the study also proposed a mathematical model to estimate electricity generation from the conversion of municipal solid waste (MSW) into methane (CH_4) in Benin. This aims to assist stakeholders in the energy sector to make appropriate decisions for a sustainable future energy transition in the country. Using local RE resources in Benin to satisfy energy demand has recently gained much interest. In view of this, several efforts have been made to investigate the Benin energy situation with potential solutions, as shown in Table 1. According to the International renewable energy agency (IRENA), only 41 % of Beninese had access to electricity at the end of 2022 [18]. Most of the rural population (around 80 %) does not have access to electricity [8] and relies on using agricultural waste and petroleum products for their energy needs. However, the country has a huge potential for RE resources that can be exploited to fill the gap between demand and supply [14].

Solar radiation ranges between 4 and 6.1 kWh/m^2/day in the Benin Republic, with a technical potential of about 3532 MW [1], and the sun shines around 2500 h per year [19]. Wind speed varies between 3 and 6 m/s in the country's coastal zones and can be exploited for small-scale wind power project development [9]. Likewise, hydropower and biomass potential can contribute to the shift towards a sustainable energy system in Benin. For instance, the waste and sanitation management company of Benin (SDGS) collects around 450,000 t/year of waste at landfill sites that can be harnessed and converted into useful energy [20]. Table 2 presents the technical potential of available RE resources in Benin. In Benin, the dynamics of electricity consumption and production reflect a steady growth in energy demand [21]. This trend stems from population expansion, ongoing economic development, and the extension of electrical services to rural regions. However, the country faces challenges related to its dependence on conventional energy sources, primarily oil and natural gas. This vulnerability exposes Benin to fluctuations in hydrocarbon prices, highlighting the urgency of diversifying the energy matrix [22]. Although renewable energies are still underutilized, promising initiatives in solar energy technologies are emerging, indicating a gradual transition towards more sustainable

Table 1
Recent studies on energy situation and electricity generation in the Benin Republic.

S/n	Country/region	Main objective	Year	Author
1	Benin	Review the energy situation in Benin	2022	[9]
2	Benin	Develop a multicriteria decision-making approach to select appropriate RE resources for electricity production	2023	[5]
3	Benin	Identified factors affecting electricity consumption in households in Benin	2022	[32]
4	Benin	Analyzed the techno-economic feasibility of installing a 10 MW grid-tied solar PV system in Benin	2023	[12]
5	West Africa	Evaluate the energy potential of selected agricultural residue in Benin	2020	[33]
6	Benin	Analyzed the techno-economic feasibility of a hybrid off-grid renewable energy system in Benin	2020	[14]
7	Benin	Analyzed and reviewed the barriers and targets for a clean energy transition in Benin	2021	[11]
8	West Africa	Analyzed Decarbonisation Strategy for Renewable Energy Integration for Electrification of West African Nations	2022	[34]
9	Benin	Analyzed the determinants of electricity consumption in Benin	2018	[35]
10	Benin	Simulated the Injection of 25 MW Photovoltaic Energy Production in the national grid of Benin	2021	[36]
11	Benin	Conducted an assessment of electricity generation from biogas in Benin	2021	[10]
12	West Africa	Reviewed strategies for sustainable energy development in West African countries	2013	[19]
13	Benin	Investigated the relationship between electricity consumption and economic growth in Benin	2020	[37]
14	Benin	Focused on electricity consumption and economic growth	2019	[38]
15	Benin	Reviewed renewable energy in the Benin Republic and its prospects	2023	[23]

Table 2
Technical potential of RE resources in Benin [1,11].

RE resources	Technical potential (MW)
Solar PV	3532
Hydropower	761
Biomass energy	749
Wind energy	322

energy sources.

To address energy and environmental challenges, the country aims to increase investments in renewable energies and promote more efficient energy practices. Solar and wind projects are being developed to harness the country's untapped renewable energy potential. Additionally, efforts are underway to strengthen collaboration between the public and private sectors, fostering the creation of strategic partnerships that can catalyze the country's energy transition. In Benin, electricity is both imported and locally produced, with the Beninese society of electricity production (SBPE) managing local generation, and the electricity company of benin (SBEE) overseeing nationwide distribution. The country's energy policy comprises key initiatives such as the Renewable Energy Policy, the solar energy promotion project (PROVES), and the renewable energy development program (PRODERE) [23]. These programs are designed to fast-track the shift toward renewable energy sources. By emphasizing sustainable practices, Benin aims to meet its energy needs while minimizing environmental impact. The adoption of informed energy policies and a focus on research and development contribute to creating an environment conducive to more sustainable electricity production and consumption in Benin. Natural gas is the first form of energy used to produce electrical energy in 2020 in Benin representing a proportion of 71.63 %. Solar PV makes up 0.30 % of the mix by form of energy whereas it was 0.74 % in 2018 (Fig. 1) [21].

Globally, the transition towards a sustainable energy system is paramount in most countries, and different studies in the literature have provided solutions to achieve a sustainable and low-cost energy transition. For instance, Oyewo et al. [24] developed a modeling approach to investigate pathways toward a fully decarbonized and least-cost electricity system for South Africa (SA). The authors found that solar PV and wind power, respectively, supplying 71 % and 28 % of the electricity demand in SA, are the best scenarios that can overcome the power sector's coal dependency. Also, Connolly et al. [25] investigated 100 % RE penetration in Ireland by 2050. The authors employed the EnergyPLAN modeling tool to assess how fluctuating RE resources can make a 100 % transition possible in the country. The results showed that 100 % RE penetration is possible in Ireland by 2050, with numerous approaches to achieve this. More importantly, it was found that the correct and adequate combination of RE resources could reduce electricity consumption and investment costs while also providing environmental benefits.

Groppi et al. [26] employed the EnergyPLAN model to investigate the optimal configurations of the Favignana Island energy system transition in 2050. The authors evaluated different energy mix scenarios with different RE penetration levels. It was found that solar PV is the appropriate technology for the transition. Lund & Mathiesen [27] analyzed the strategies for 100 % RE integration in Denmark for 2030 and 2050. The results showed that RE integration with local resources is feasible in the country but very complex to achieve when considering the availability of biomass resources to be integrated. Ćosić et al. [28] applied the EnergyPLAN model to investigate 100 % RE penetration in Macedonia by 2050. The authors found that 50 % RE integration is more feasible by 2050 than the 100 % target in the current situation in the country. This is because high biomass penetration would be difficult to achieve in Macedonia.

Graça Gomes et al. [29] employed the EnergyPLAN model to investigate energy system scenarios with a high RE fraction in Portugal. It was observed that the RE sources could meet the electricity demand for the country while maintaining interconnection with neighboring countries. The authors suggested that an installed capacity mix with more wind power is the most favorable scenario for decarbonizing the Portuguese electricity sector. Similarly, Icaza-Alvarez et al. [30] applied EnergyPLAN to analyze strategies for the penetration of 100 % RE by 2050 in the Ecuadorian Amazon. The authors found that the energy mix combining 33.04 % solar PV, 36.47 % hydropower, and 29.73 % wind power is the most appropriate combination to achieve 100 % RE by mid-century. Izanloo et al. [31] investigated energy development plans to maximize RE share in the Mazandaran region. The authors utilized a combination of RE sources such as wind, biomass, solar PV, and hydropower.

The studies reviewed above indicate countries' willingness and interest to increase the share of RE resources in their total energy mix in the future for a sustainable energy transition. However, only a few studies have been conducted in Africa to integrate 100 % RE by mid-century. To the authors' best knowledge, this study is the first to be conducted in the Benin Republic. It aims to develop RE scenarios, integrate RE sources into the country's supply mix, and contribute to its Nationally Determined Contribution (NDC) under the Paris Agreement. The study's main objective is to investigate options that could help

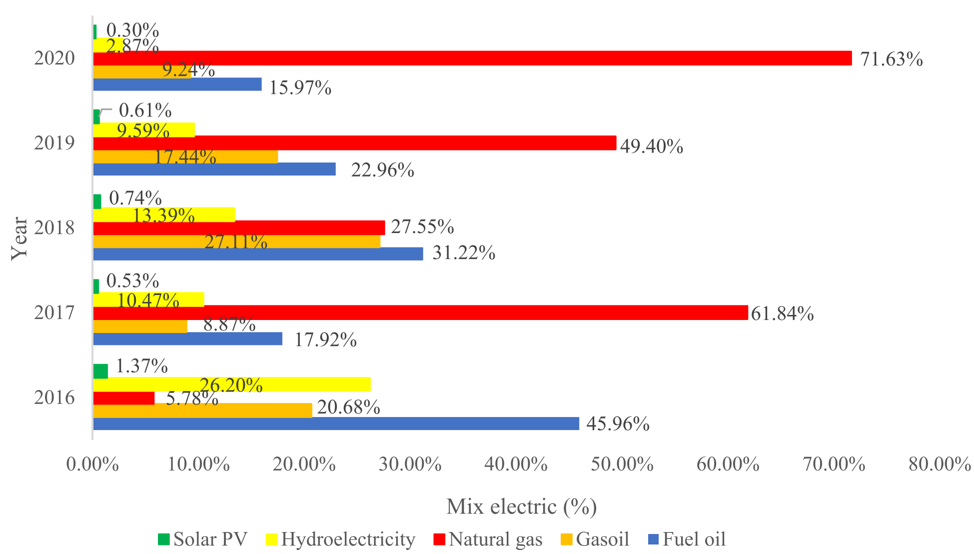

Fig. 1. The mix electric of Benin from 2016 to 2020 [21].

Benin achieve its national RE penetration targets by mid-century. The study findings are useful to decision-makers and stakeholders in the energy sector to develop appropriate policies that would promote the development of RE technologies for Benin's socio-economic development. The study's main contributions are as follows:

(1) develop three scenarios to analyze strategies to increase RE in the supply mix. The scenarios include the government targets scenario, the 2 % RE per year scenario, and the 50 % by mid-century scenario.
(2) propose a mathematical approach to estimate electricity generation from MSW abundantly available in Benin.

The remaining study sections are presented as follows: Section 2 presents the materials and methods used for developing the energy mix scenarios. Section 3 presents the main results and discussions, while Section 4 provides concluding remarks.

2. Methodology

This section presents the material and methods used in the study to propose electricity supply scenarios by integrating renewable energy sources to meet national targets by 2050.

2.1. Study location and electricity demand data

Benin Republic is located in West Africa, in the Gulf of Guinea of the Atlantic Ocean. The country is bothered by the Republic of Niger to the north, Burkina Faso to the northwest, Nigeria to the east, and the Republic of Togo. It lies within latitude 6°29′ north and longitude 2°36′ west, as shown in Fig. 2 [9,16]. The country's surface area is estimated to be 112,763 km^{2,} with a population of 13.3 million [9]. The official language of Benin is French, and Porto-Novo is the capital city. However, Cotonou city is the economic hub with around 700,000 inhabitants [16]. Solar irradiation increases gradually from the southern part of the country to the northern part. Favorable wind potential exists in coastal areas of Benin that can be harnessed for small-scale wind power projects [23].

Fig. 3 shows the electricity demand profile of Benin. It can be seen that the electricity demand will increase from 1.06 TWh in 2015 to 3.74 TWh in 2030. A recent report by SIE-Benin [21] indicates that the country's electricity mix was dominated by natural gas (72 %), while non-biomass RE resources (Solar PV and Hydropower) represented 3.2 % of the energy mix in 2020.

Based on the electricity demand projection of 2030 by [1], the electricity demand of 2050 is estimated using eq (1) [40]:

$$E_{2050} = E_{2030}\left(1 + \frac{r}{100}\right)^{n} \tag{1}$$

Where r is the average population growth rate of Benin ($r = 2.7$ %), E_{2030} and E_{2050} are the electricity demand in 2030 and 2050 respectively, and n is the number of years ($n = 20$). The estimated electricity demand for 2050 is presented in Table 5.

2.2. RE modeling scenarios approach

This study considers RE technologies, including solar PV, concentrated solar power (CSP), hydropower, and wind energy, for developing

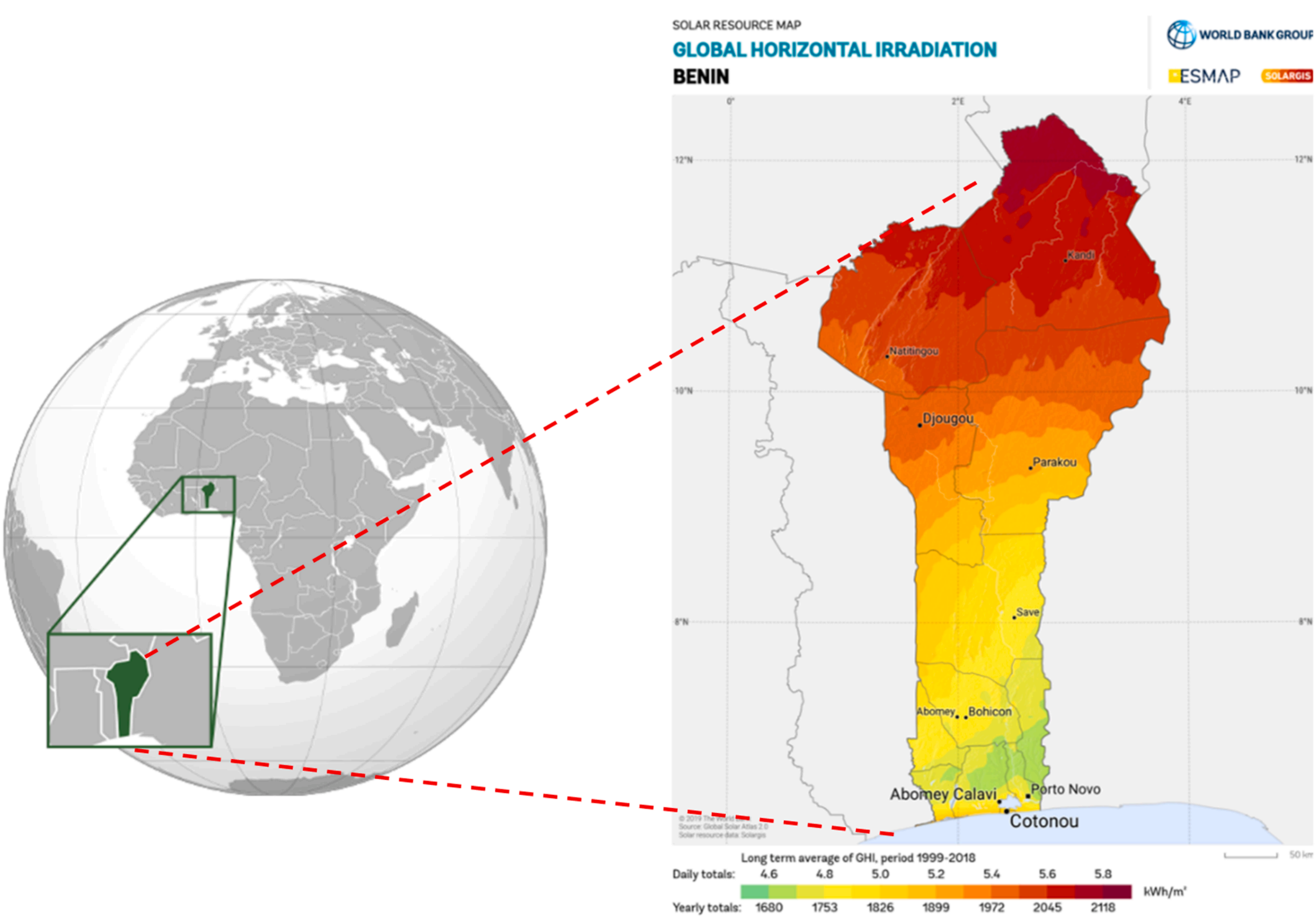

Fig. 2. Benin's geographical location, right: in West Africa location [16], left: Benin's solar resource map [39].

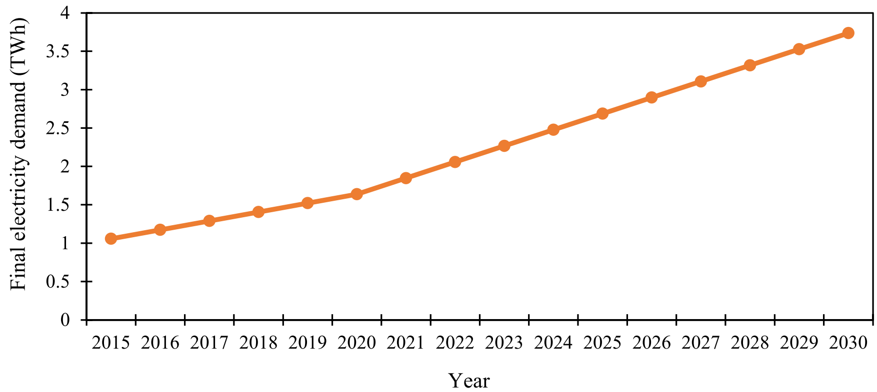

Fig. 3. Final electricity demand projection for Benin (TWh) [1].

the scenarios. The study considered these RE technologies because they are mature technologies fully installed in different countries. Moreover, a recent study by Akpahou & Odoi-Yorke [5] alluded to the fact that these RE technologies are the most promising alternatives, with solar PV being the best technology in Benin. Notwithstanding, natural gas (NG) currently dominates Benin's primary energy. NG is an abundant, secure, and flexible energy source. Compared with some other fossil fuels, NG emits the least amount of carbon dioxide (CO_2), making it the cleanest burning fossil fuel of all. It is currently imported from Nigeria for electricity generation in Benin, constituting more than 72 % of the supply mix [9]. Hence, NG, a non-renewable source, is considered in the study scenarios due to its low emissions, flexibility, and maturity of natural gas power plants. The capacity factor is one of the most reliable indicators for assessing power plant production performance [41]. According to IRENA [1], the capacity factor for Benin's solar PV and wind power plants are 18.8 and 12.6% respectively. The hydropower capacity factor ranges between 10 and 99 % depending on water availability at the location [41]. Table 3 presents the RE technology capacity factors considered in the study. The next subsection highlights the EnergyPLAN model utilized in this study.

2.2.1. *EnergyPLAN model*

The EnergyPLAN model is a computer-based model for energy system analysis developed to assist in designing energy planning strategies on national, local, and/or regional scales [42]. EnergyPLAN is more efficient for long-term energy planning strategy analysis and suitable for RE integration plans. It is an input and output energy model developed at the University of Aalborg and can perform technical, economical, and environmental emissions analyses of power plants. It has been employed in countries like Denmark, Macedonia, Nigeria, and South Africa [25,30,41]. EnergyPLAN is used in this study to analyze RE resource integration strategies in Benin up to 2050. The analysis also included solving the insufficient electricity supply in Benin based on critical excess electricity production (CEEP) and power production importation (PPI).

EnergyPLAN inputs data include energy demands, renewable energy sources, capacities of energy plants, costs, and various optional simulation strategies, with a focus on import/export dynamics and surplus electricity production. The outputs generated include energy balances, annual production outcomes, fuel consumption details, import/export figures, and comprehensive cost analyses, incorporating income derived from electricity exchange. It can be used for Technical analysis, Market exchange analysis, and feasibility studies [42]. Various technologies can be incorporated to comprehensively reconstruct all components of an energy system, facilitating the analysis of factors such as wind integration technologies and the intricate relationship between electricity and heat supply, especially in scenarios with high levels of combined heat and power (CHP) [43]. The model allows for the implementation of diverse regulation strategies, with a focus on heat and power supply, import/export dynamics, ancillary services, grid stability, and surplus electricity production. Operating on an annual basis with an hourly time step, the model conducts a thorough analysis of an entire smart energy system, considering factors such as losses and other technical intricacies in simulating a renewable energy (RE) system [25]. Its applicability extends to district cooling and heating analysis, electric vehicle integration, RE integration, and electricity import/export scenarios. Offering flexibility, the model accommodates various energy strategies for comprehensive energy system analysis, relying on analytical programming rather than dynamic, advanced mathematical tools, or iterative programming approaches [41].

The flowchart showing the optimal simulation model process adopted in this study is shown in Fig. 4. Optimal modeling in this study is for a technology or combination of technologies that meet the demand with low cost, minimal or no CEEP and low emissions. This model prioritizes the integration of several available RE sources. In this study, the NG supply is first provided to meet the demand, and the other scenarios are based on a combination of NG capacity and RE technologies. After every simulation, different outcomes based on the CEEP or PPI are expected. The CEEP-only warning indicates excess electricity generation from RE sources that must be reduced. On the other hand, only the PPI warning indicates that the power produced is insufficient to cover the demand and, therefore, supply must be increased. Also, no CEEP and PPI warning mean supply is sufficient to meet demand. In contrast, the CEEP and PPI warnings mean that the power production is insufficient to meet the demand, and there is excess production from some RE technologies. Thus, action should be taken to adjust accordingly. In addition, when the RE % is less than the required to meet the target, RE% is increased, if not, if RE % is greater than the required, it is decreased. In the case where RE % is equal to the required but there is still CEEP error, the technology combination with the lowest CEEP is chosen.

2.2.2. *Mathematical modeling of RE technologies*

EnergyPLAN uses mathematical equations to estimate power output from RE technologies such as wind power, solar PV, and hydropower plants. To estimate the power the wind farm generates, we need the hourly wind data profile for Benin. This study utilized the 10 m wind speed data downloaded at latitude 11.022 and longitude 2.610, with an average wind speed of 1 to 6 m/s, as shown in Fig. 5. The maximum wind turbine power output is computed using Eq. (2) [44]:

Table 3
Capacity factor of RE plants.

RE Technology	Capacity factor (%)	Source
Solar PV	18.8	[1]
Wind	12.6	[1]
Hydropower	45	[41]
CSP	26	[41]

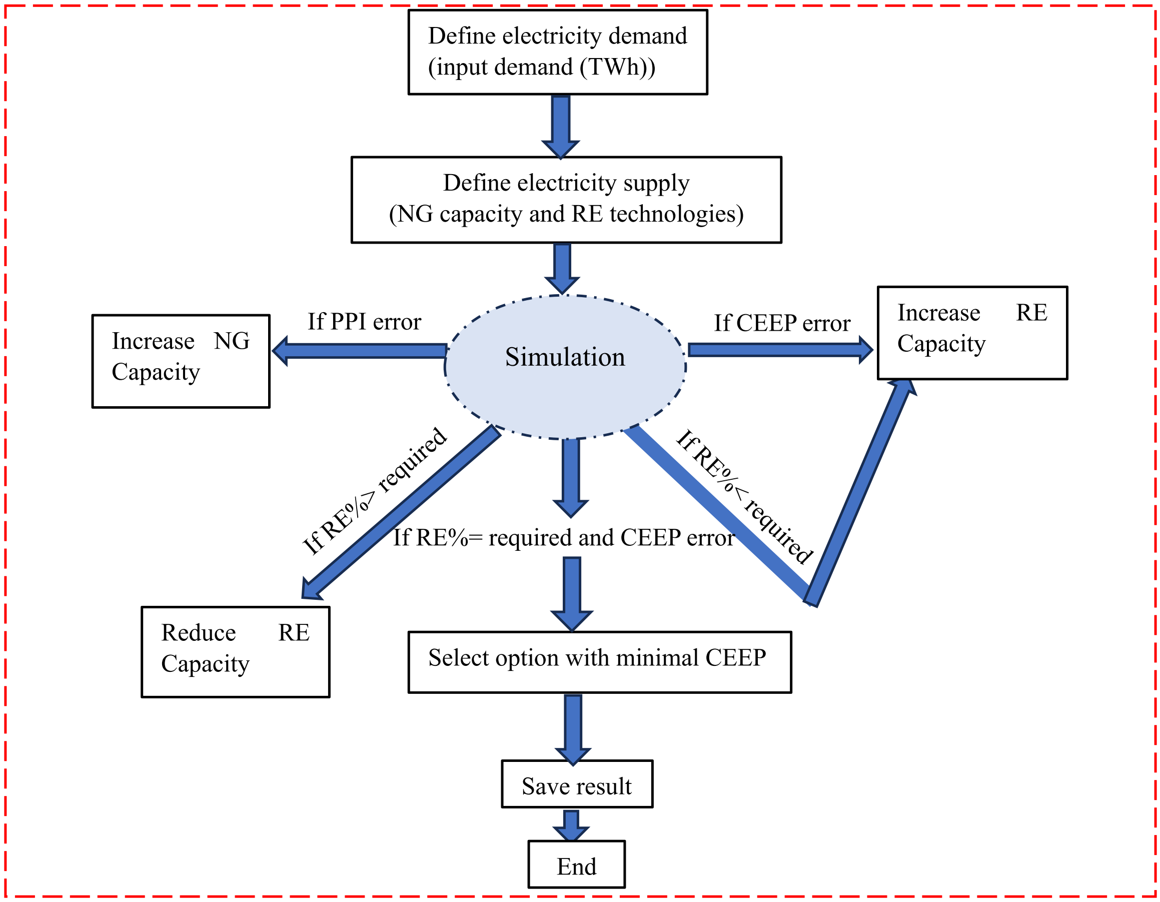

Fig. 4. Flowchart of optimal simulation model.

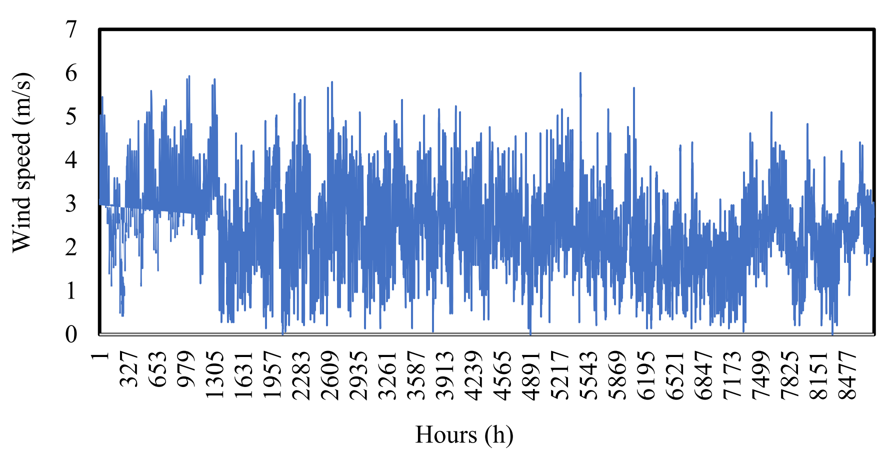

Fig. 5. Wind speed (m/s) distribution profile for Benin at 10 m hub height.

$$P_w = \frac{1}{2} \times \eta \times C_p \times \rho \times A \times V^3 \tag{2}$$

Where C_p is the power coefficient, η is the combined transmission and electrical efficiency, ρ is the air density (kg/m^3), A is the cross-sectoral area of the rotor (m^2), and V is the wind speed (m/s).

Fig. 6 shows Benin's hourly solar insolation distribution profile obtained at latitude 11.022 and longitude 2.610. The northern part of Benin is more favorable for developing solar energy technologies since it receives the maximum solar insolation throughout the year [12]. The hourly output power from a solar PV plant is estimated using Eq. (3) [45]:

$$P_{PV} = H(t)_c \times S \times \eta_{PV} \tag{3}$$

With $H(t)_c$ is the solar insolation (kW/m^2), S the module surface area (m^2), and η_{PV} the efficiency of the solar module.

The power output from a hydropower plant is generally estimated as follows [34]:

$$P_{HY} = Q \times H \times \rho \times \eta \times g \tag{4}$$

Q is the water flow rate (m^3/s), η is the generator's efficiency (%), ρ represents the water density (kg/m^3), H is the fall height (m), and g is the acceleration of gravity (m/s^2). The main hydropower source of Benin

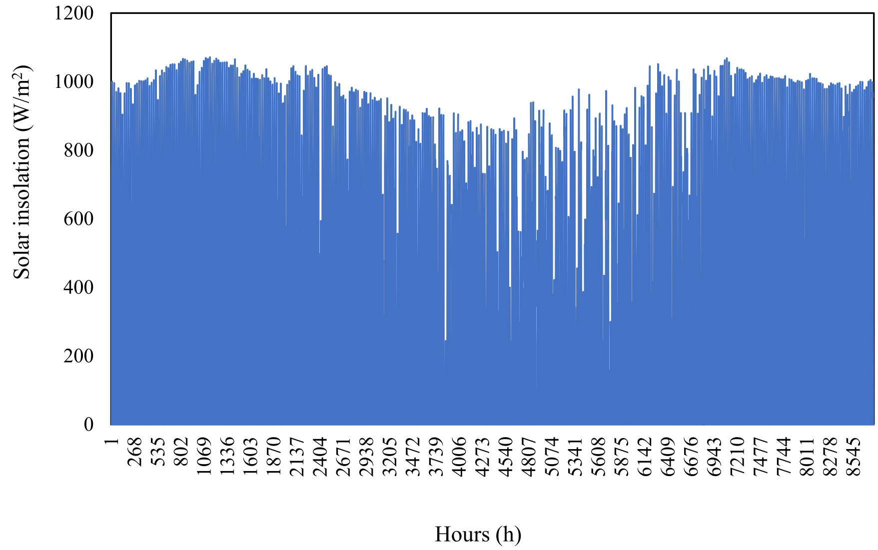

Fig. 6. Hourly solar insolation (W/m^2) distribution profile for Benin.

is the Nagbeto hydropower plant which produces electricity for both Benin and Togo. Electricity is also imported from Akonsombo hydroelectricity power in Ghana and Nigeria. For this study, a typical hydropower run-over river profile is used as illustrated in Fig. 7.

The energy output of the CSP system is the amount of electricity the system produces over a period of time. It depends on factors such as the solar resources available, the optical and conversion efficiency, and the system's availability. The potential outpower P_{CSP} calculation process is described as follows [42]:

First, the demand for solar thermal S_{CSP} is estimated:

$$S_{CSP} = \frac{C_{CSP}}{\mu_{CSP}} \tag{5}$$

Then, the program estimates the maximum output production P_{CSP} as follows:

$$\left\{ \begin{array}{c} If\ Q_{CSP} > S_{CSP},\ then\ P_{CSP} = S_{CSP} \times \mu_{CSP} \\ If\ Q_{CSP} < S_{CSP},\ then\ P_{CSP} = (Q_{CSP} + storagecontent) \times \mu_{CSP} \\ If\ P_{CSP} > C_{CSP},\ then\ P_{CSP} = C_{CSP} \end{array} \right\} \tag{6}$$

Where C_{CSP} is the capacity of the CSP plant, μ_{CSP} is its efficiency and Q_{CSP} its annual solar thermal input. Further details are available in the EnergyPLAN documentation version 16.2 [42].

EnergyPLAN estimates renewable factors (R_f) to evaluate the prospects of RE energy systems. Eq. (7) is applied to compute the renewable factors [30]:

$$R_f = \left(1 - \frac{\sum P_{non-renewable}}{\sum P_{renewable}}\right) \times 100 \tag{7}$$

Where $\sum P_{renewable}$ represents the sum of energy output from renewable energy sources and $\sum P_{non-renewable}$ is the sum of energy generated from non-renewable sources such as natural gas.

Furthermore, Table 4 outlines the investment costs in the EnergyPLAN Cost Database for the different technologies considered for this study. The EnergyPLAN cost database provides data for different years including 2015, 2020, 2030, and 2050 that enable users to perform accurate simulations and analysis [46]. The economic metrics considered include investment cost, operation, and maintenance costs (O&M), power plant lifetime, and fuel costs. Solar PV has the lowest investment and O&M costs among all technologies (Table 4). The emissions generation from non-renewable technologies is evaluated, with an emission factor of 56.9 kg/GJ used for natural gas plants [41].

2.2.3. *Modeling electricity generation from the conversion of municipal solid waste into methane in Benin*

To determine the contribution of biomass energy resources, this study estimated the electricity generation from converting municipal

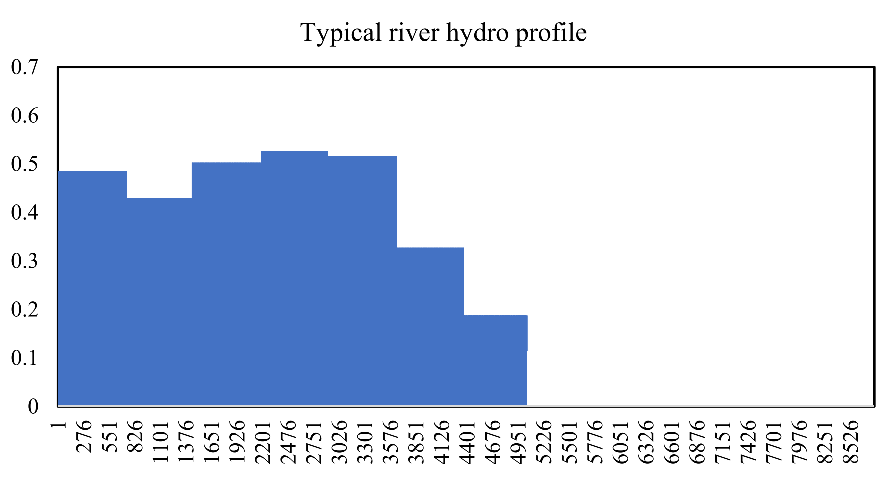

Fig. 7. Hydropower plant profile used in the study [41].

Table 4
Techno-economic parameters for power generation technologies [34,41,46].

Year	Type of power plant	Investment cost (million USD/ MWe)	Lifetime (years)	Fixed O&M (% of investment)	Fuel cost (USD/ GJ)
2025	NG	0.99	27	3.12	9.1
	Solar PV	0.76	35	1.31	
	CSP	4.15	30	4	
	Wind	0.95	27	3.2	
	Hydro	5.61	60	1.5	
2030	NG	0.98	27	3.16	10.2
	Solar PV	0.68	40	1.28	
	CSP	3.8	30	4	
	Wind	0.91	30	3.27	
	Hydro	5.62	60	1.5	
2050	NG	0.9	27	3.26	12.2
	Solar PV	0.56	40	1.32	
	CSP	3.4	30	4	
	Wind	0.93	30	3.4	
	Hydro	5.62	60	1.5	

solid waste (MSW) available in Benin into methane (CH_4). According to World Bank [47], MSW in Benin is estimated to amount 0.75 kg/capita per day in 2025. Organic and paper constitute the highly degradable class (55 %) while plastics (7 %) are in the moderately degradable class of MSW waste classification according to World Bank [48].

A similar method using eq (1) is used to estimate the waste production W_i (kg/capita/day) in 2030 and 2050 based on World Bank [47].

According to Silva dos Santos et al. [49], the methane gross energy production from the landfills is calculated using Eq. (8):

$$E_{CH4} = 365 \times \frac{L_O \times f_{LF} \times N \times q \times \varepsilon \times \varepsilon_{gas}}{1000} \times L_{CH4} \tag{8}$$

L_O is the biogas production factor of MSW (m^3/t), f_{LF} is the fraction of MSW sent to landfills, ε_{gas} is the gas collection efficiency, N is the urban population, q is the quantity of waste collected (kg/capita/day), ε is the waste collection efficiency, and L_{CH4} is the lower heating value of methane taken as 35.5 MJ/m^3 [50].

In Benin, the MSW collection efficiency (ε) is projected to reach 50 % by 2025 [51]. In addition, for the average potential of CH_4 calculation from landfills, a collection efficiency (ε_{gas}) of 55.5 % was considered [49]. The urban population of Benin is projected to reach 7050,785; 8562,802 and 18,173,777 in 2025, 2030, and 2050 respectively [52]. An average value of CH_4 production factor of 170 m^3/t and 262.5 m^3/t was used for moderately degradable and highly degradable waste, respectively, as suggested by the World Bank [48].

The annual electricity generation from E_{CH4} is estimated using Eq. (9) using a power plant internal combustion efficiency of 35 % and a capacity factor of 80 % [48,50].

$$E_{el} = 0.35 \times 0.8 \times E_{CH4} \tag{9}$$

2.3. Description of the developed scenarios

2.3.1. Government targets scenario (Base scenario)

The government targets scenario, known as the base scenario is the reference scenario considering the government targets to integrate more RE resources by mid-century. The significance of integrating RE has surged to the forefront of Benin's domestic agenda and the broader West African landscape. Notably, the ECOWAS region has rallied around an ambitious goal of achieving a robust 48 % RE penetration by the culmination of 2030 at the regional level [1]. Furthermore, Benin has collaboratively engaged with other ECOWAS nations to implement the SEforALL country action plan, which involves formulating renewable energy action plans. The specific objective is to achieve a 24.6 % penetration of renewable energy in Benin's energy mix by 2025 [8]. Additionally, Benin has set its national target for renewable energy integration at 44 % by 2030, alongside the goal of achieving universal electricity access [17]. Moreover, as most long-term energy transition plans end dates currently go beyond 2030 [53], including 2050 for the Gambia [54], 2060 for Ghana [55], and 2050 for Nepal [56], this study also investigated the strategies towards RE integration in 100 % electricity access by 2050 in Benin. The government targets for 2025 and 2030 indicate 19.4 % of RE integration in 5 years. This shows that Benin would reach 100 % RE integration by 2050. The targets for 2025, 2030, and 2050 are considered under this scenario. The study intends to propose an optimal energy mix that propels the nation towards these ambitious targets and lays the groundwork for a sustainable energy future. Table 5 presents RE targets and the electricity demand data.

2.3.2. 2 % RE integration per year scenario

This scenario is built on the government targets scenario and assumes a 10 % RE every 5 years based on 2 % per year integration basis. Considering the 3.2 % of RE in Benin's supply mix as of 2020 [21], RE integration would be 23.2 % and 63.2 % in 2030 and 2050 respectively under this scenario.

2.3.3. 50 % RE integration by mid-century scenario

This scenario is also built on the government targets scenario but considers a more realistic RE integration percentage based on the country's current situation. It assumes that, 50 % of electricity comes from RE sources and 50 % comes from natural gas. In a country like Denmark, 50 % of RE by 2050 is found to be more realistic than 100 % by the same period.

3. Results and discussions

This section presents and discusses the main findings from the study by analyzing different scenarios to meet government targets by 2025, 2030, and 2050. In addition, electricity generation from MSW and policy implications are discussed.

3.1. Government targets scenario analysis

3.1.1. Supply mix analysis to meet 2025 RE target in the supply mix under government targets scenario

Benin is projected to have a significant electricity demand in 2025, estimated at 2.7 TWh/yr. However, the current installed capacity of 181 MW is insufficient to meet this demand. This reveals a substantial energy supply gap that needs to be addressed urgently to ensure the country's energy security. To bridge the demand-supply gap, it is estimated that a natural gas (NG) power plant with a capacity of 474 MW will be needed (Fig. 8). This represents a substantial increase in power generation capacity and highlights the need for significant infrastructure development in the energy sector. The natural gas power plant is expected to emit approximately 1.225 $MtCO_2$. This raises environmental concerns, as increased carbon emissions can contribute to climate change and adversely affect the environment and public health. Therefore, mitigating these emissions should be a priority, potentially through cleaner energy alternatives or carbon capture and storage technologies. The construction and operation of the NG power plant require extensive financial investments. With a total annual investment cost of USD 469 million and an annual operating cost of USD 237 million, there is a need for a clear financial plan and sources of funding to support the

Table 5
RE targets and electricity demand data for Benin [1,8].

Target year	2025	2030	2050
RES share (%)	24.6	44	100
Electricity demand (TWh/yr)	2.7	3.74	6.4
RE electricity production (TWh/yr)	0.66	1.63	6.4

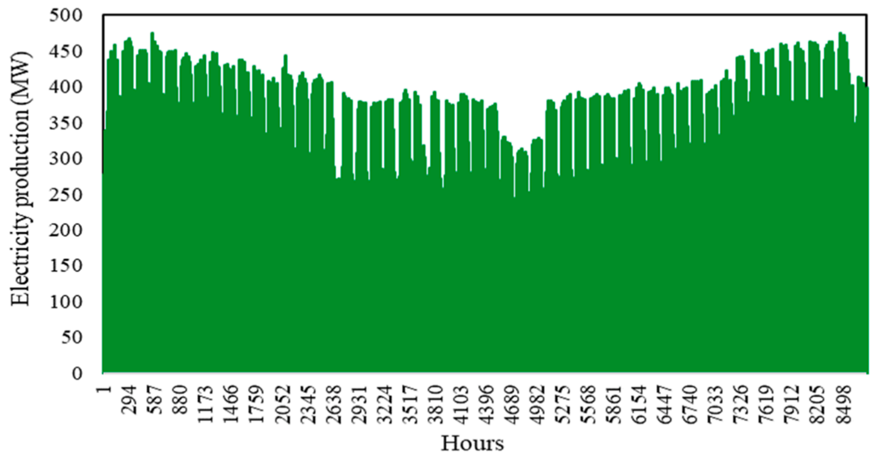

Fig. 8. Hourly electricity production from natural gas power plant.

development of the power plant. This could involve public-private partnerships or seeking international funding assistance.

The current situation, where peak power production is only a fraction of the projected demand, can result in frequent power shortages, affecting the country's economic growth and quality of life for its citizens. Increasing capacity through the NG power plant is essential to enhancing energy security and reliability.

It is noteworthy that Benin possesses a significant hydropower potential of 769 MW. This can potentially replace NG and reduce external energy dependence, provided the necessary investments and societal impacts are thoroughly investigated. Fig. 9 shows the proposed energy mix ccombinations' annual costs, investment costs, CO_2 emissions, and RES electricity production for 2025. It can be observed in Fig. 9 that RES in Benin can sustainably contribute to the energy transition if adequate investments are made. To achieve 24.6 % of RE in the total mix by 2025, a hydropower capacity of 169 MW complementing NG would be sufficient with no CEEP generation. This would reduce emissions from 1.225 $MtCO_2$ to 0.923 $MtCO_2$ with a total investment and total annual cost of USD 1.303 billion and USD 237 million, respectively.

It can also be observed in Fig. 9 that the NG+PV scenario appears to be the most economically feasible option, with investment and total annual costs of USD 591 million and USD 203 million, respectively. This requires a solar PV capacity of 311 MW, sufficient to meet the 24.6 % targets with the generation of 38 MW of CEEP. Moreover, the available technical wind potential is about 322 MW in Benin, slightly higher than the required 303 MW to meet the target. Therefore, combining NG with 100 MW of wind and 209 MW of PV capacity would be a feasible option to meet the target (Table 7). This installation would require a total investment and annual cost of USD 609 million and USD 206 million, respectively. The demand and supply for RE integration in this scenario are illustrated in Fig. 10, with wind and solar PV generating 0.22 TWh and 0.44 TWh/year with no CEEP. The non-uniform electricity production observed reflects the solar PV and wind distribution profiles of the country. Solar resources are higher in northern Benin, adequate for deploying solar technologies, whereas wind is more available in the coastal areas appropriate for small-scale wind plant development [23].

The last considered option is the combination NG+PV+Wind+Hydro. This combination would require a total investment and annual cost of USD 870 million and USD 218 million, respectively, with an installed capacity of 130 MW of PV, 60 MW of wind, and 65 MW of small hydro. Fig. 11 presents the demand and supply mix graph, with solar PV, wind, and hydro generating 0.28, 0.13, and 0.25 TWh/yr, respectively. This combination is the most feasible with no CEEP and the least investment cost considering PV, Wind, and hydro sources.

3.1.2. Combinations analysis to meet 2030 and 2050 RE targets in the government targets scenario

Figs. 12 and 13 illustrate the different combinations developed to meet 2030 and 2050 targets with investment and total annual costs. In total, 22 combinations have been considered, starting from NG only to the supply mix of NG+PV+Wind+Hydro+CSP. To meet the 3.74 TWh of electricity demand, an installation capacity of 657 MW of NG is required. This would result in an emission of 1.696 $MtCO_2$ and a total investment and annual cost of USD 644 million and USD 361 million, respectively. Similarly, an installed capacity of 1124 MW of NG is

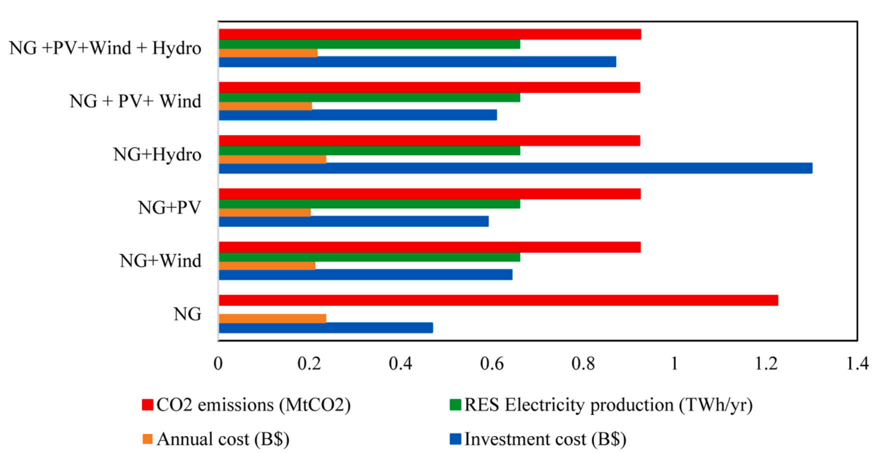

Fig. 9. Annual and investment cost, CO_2 emissions, and RE Electricity production from different sources.

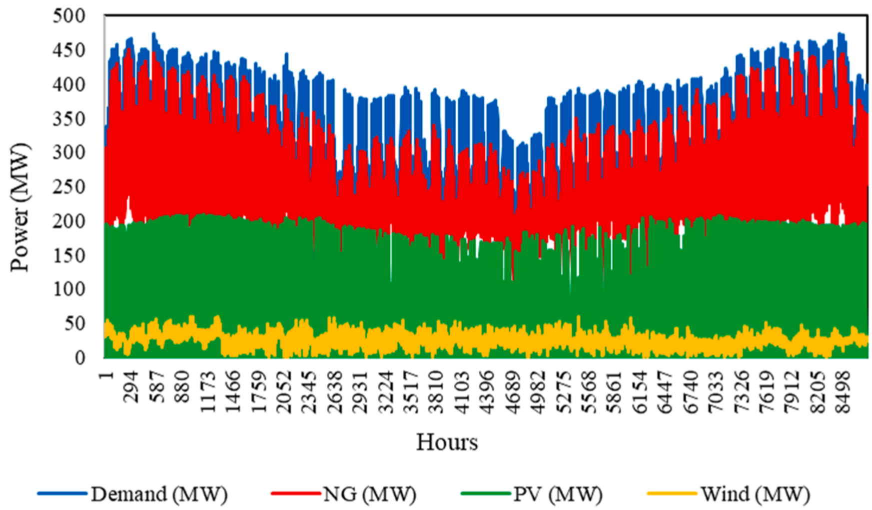

Fig. 10. Hourly electricity demand and supply for 24.6 % optimal RES integration using NG+PV+Wind.

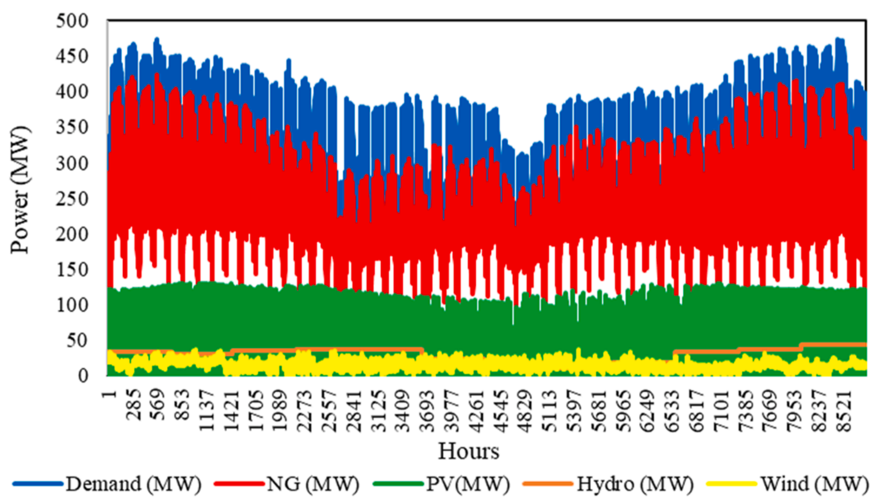

Fig. 11. Hourly electricity demand/supply for 24.6 % optimal RES integration using NG+PV+Wind +Hydro.

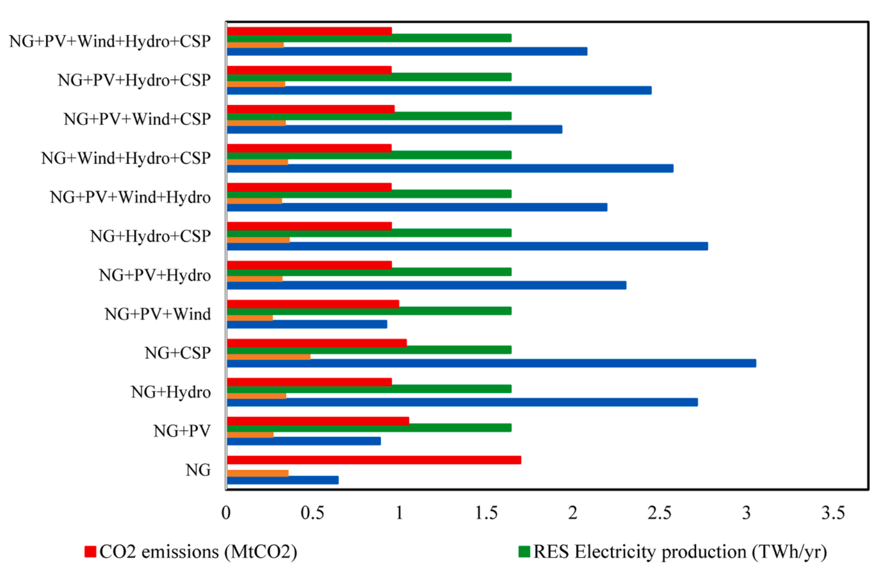

Fig. 12. Annual and investment cost, CO_2 emissions, and RE Electricity production sources under government targets scenariofor 2030.

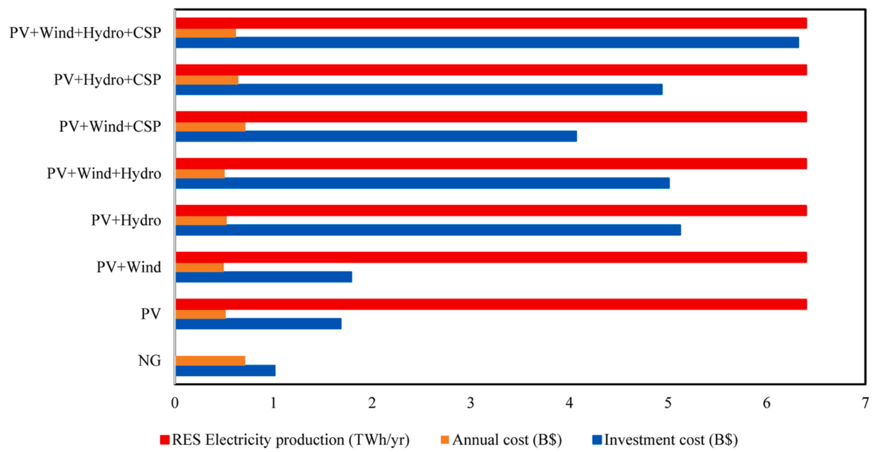

Fig. 13. Annual and investment cost, CO_2 emissions, and RE Electricity production under government targets scenario for 2050.

required to provide 100 % renewable electricity to Beninese by 2050. The total CO_2 emission is estimated to be 2.9 $MtCO_2$ with an investment cost and total annual cost of USD 1012 million and USD 713 million respectively.

Also, a solar PV capacity of 770 MW and 3002 MW is required to achieve 44 % and 100 % of RE by 2030 and 2050 respectively. However, this would generate 392 MW and 2349 MW of CEEP as presented in Table 6, and require a total investment cost of USD 524 million and 1681 million respectively. In addition, CO_2 emissions would decrease to 1.05 $MtCO_2$ in 2030 and 0.0 $MtCO_2$ in 2050 in 100 % RE situation. It is important to consider the integration of CSP technologies in the long term in Benin with the abundant solar insolation received throughout the year [5]. The NG+CSP combination would require an installation capacity of 708 MW of CSP plants to complement NG and meet the 44 % RE target by 2030. The RE share in the primary energy supply (PES) is about 24.5 %, as presented in Table 6, with a generation of 330 MW of CEEP. The total investment cost and annual cost are USD 3.05 billion and USD 487 million, respectively.

Similarly, an installation capacity of 2750 MW of CSP is required to have 100 % of RE in the final electricity supply mix of Benin by 2050 with an investment cost of USD 9.35 billion. Despite CSP's higher installation costs compared to wind and solar PV, the benefits are substantial in the long run. According to [57], the average levelized cost of electricity for CSP plants fell by 69 % in 2022, decreasing from USD 0.380 /kWh in 2010 to USD 0.118 /kWh in 2022. This demonstrates the cost-effectiveness of CSP plants, and governments could invest in CSP projects to develop their future energy system.

As depicted in Fig. 12, the most economically feasible combinations in 2030 are NG+PV and NG+Wind+PV, with a total investment cost of USD 887 million and USD 924 million, respectively. Fig. 14 also illustrates the investment cost, CO_2 emissions, and RE Electricity production for all combinations in 2050. Utility-scale solar PV and onshore wind plant installation costs have decreased in recent years. According to IRENA [57], the LCOE of utility-scale PV plants fell by 89 % between 2010 and 2022, from USD 0.445 /kWh in 2010 to USD 0.049 /kWh in 2022. Similarly, the LCOE for onshore wind power plants has

Table 6
Summary of the outputs for different considered options under the government targets scenario.

Year	Combinations	NG (MW)	PV (MW)	Wind (MW)	Hydro (MW)	CSP (MW)	RE Electricity production (TWh/yr)	RE share in electricity mix (%)	RE in PES (%)	CEEP (MW)
2025	NG only	474	–	–	–	–	–	–	–	–
	NG+PV	358.2	311	–	–	–	0.66	24.6	12.8	38
	NG+Wind	358.2	–	303	–	–	0.66	24.6	12.8	0
	NG+Hydro	358.2	–	–	169	–	0.66	24.6	12.8	0
	NG+PV+Wiind	358.2	209	100	–	–	0.66	24.6	12.8	0
	NG+PV+Wind+Hydro	358.2	130	60	65	–	0.66	24.6	12.8	0
2030	NG only	657	–	–	–	–	–	–	–	–
	NG+PV	370.4	770	–	–	–	1.64	44	24.1	392
	NG+Hydro	370.4	–	–	418.4	–	1.64	44	24.1	0
	NG+CSP	370.4	–	–	–	708	1.64	44	24.5	330
	NG+PV+Wind	370.4	592.4	174	–	–	1.64	44	25.3	263
	NG+PV+Hydro	370.4	174	–	324	–	1.64	44	26.1	0
	NG+Hydro+CSP	370.4	–	–	348	119.2	1.64	44	26.1	0
	NG+PV+Wind+Hydro	370.4	139.2	87	294.5		1.64	44	26.1	0
	NG+Wind+Hydro+CSP	370.4	–	87	308.5	104.4	1.64	44	26.1	0
	NG+PV+Wind+CSP	370.4	321	176	–	248	1.64	44	26.1	247
	NG+PV+Hydro+CSP	370.4	124	–	322	50	1.64	44	26.2	0
	NG+PV+Wind+Hydro+CSP	370.4	124	87	276	45	1.64	44	25.4	0
2050	NG only	1124	–	–	–	–	–	–	–	–
	PV	–	3002	–	–	–	6.4	100	45.8	2349
	CSP	–	–		–	2750	6.4	100	46.3	2100
	PV+Wind	–	2695	300	–	–	6.4	100	28.1	2071
	PV+Hydro	–	1620	–	750	–	6.4	100	62.3	1356
	PV+Wind+Hydro	–	1400	300	702	–	6.4	100	65.9	1180
	Wind+Hydro+CSP	–	–	300	740	1218	6.4	100	67.6	1045
	PV+Wind+CSP	–	1800	300	–	818	6.4	100	48.8	1995
	PV+Hydro+CSP	–	1183	–	750	400	6.4	100	62.5	1320
	PV+Wind+Hydro+CSP	–	1000	300	600	538	6.4	100	66.3	1262

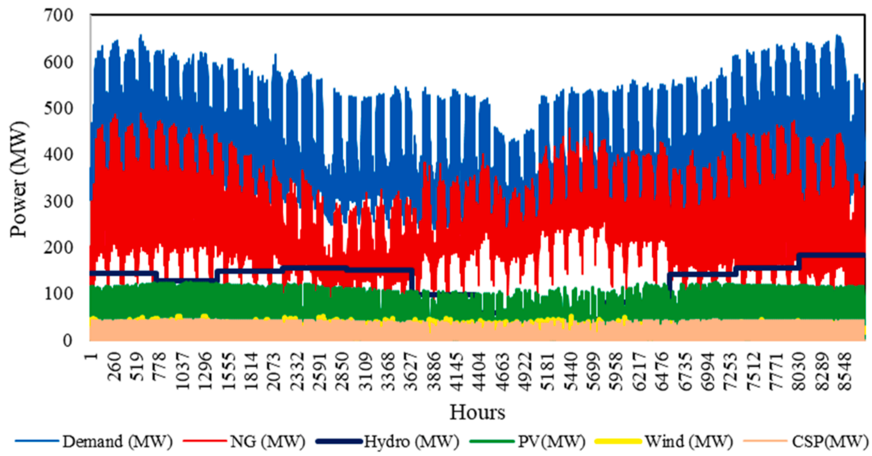

Fig. 14. Hourly electricity demand/supply for 44 % optimal RES integration using NG+PV+Wind +Hydro + CSP.

experienced a notable decline, plunging from USD 0.107/kWh in 2010 to an impressively low USD 0.033 /kWh in 2022. These trends highlight renewable energy investments' compelling economic prospects in the scenarios discussed.

Furthermore, the combination including NG, PV, wind, hydro, and solar CSP to provide 44 % of RE by 2030 would include 124 MW of PV, 87 MW of wind, 276 MW of hydro, and 45 MW of CSP. This would require an investment cost of USD 2.077 billion to reach the target with no CEEP. This option's demand and supply mix is illustrated in Fig. 12, with an electricity generation of 0.26 TWh from solar PV, 0.19 TWh from wind, 1.08 TWh from hydro, and 0.1 TWh from CSP per year. Moreover, an optimal mix of 600 MW of Hydro, 1000 MW of PV, 300 MW of Wind, and 538 MW of CSP would provide 100 % of RE and 100 % of electricity access in Benin by 2050. The hourly electricity production from this combination is illustrated in Fig. 15 and the system would require a total investment cost of USD 6.0 billion. The required capacity and electricity production from the different options are presented in Table 6. Benin could make substantial progress toward mitigating the adverse effects of climate change by transitioning to renewable energy sources and reducing carbon emissions. Due to its geographical location, Benin receives a lot of solar radiation, with the sun shining for more than 2500 h a year. The technical potential of solar at 3532 MW is not much exploited yet and needs more investigation to boost CSP and solar PV deployment in Benin.

3.2. 2 % RE growth per year scenario analysis

This scenario considers an annual RE integration rate of 2 % and is built up on the government scenario. This implies that by 2050, Benin's electricity supply would contain 63 % of RE. The estimated RE electricity generation for 2050 would reach 4.04 TWh, comprising a well-balanced mix of 414 MW from natural gas, 410 MW from solar photovoltaic (PV), 300 MW from wind, 600 MW from hydropower, and 65 MW from concentrated solar power (CSP). This strategic combination not only diversifies the energy portfolio but also contributes to environmental goals. It necessitates an annual cost of USD 554 million and a total investment cost of USD 4.5 billion, slightly lower than in the government targets scenario for the same period. The breakdown of energy generation indicates that solar PV contributes 0.87 TWh, wind energy 0.66 TWh, hydropower 2.36 TWh, and CSP plants 0.15 TWh. While the RE integration target in this scenario is marginally lower than the government's objectives, it results in reduced investment costs. It makes the attainment of 63.2 % by 2050 more achievable for the government compared to the more ambitious 100 %. The experiences of countries like Ireland, Denmark, and Macedonia, where achieving 100 % RE integration has proven challenging, underscore the practicality of opting for a slightly lower integration rate by mid-century, which remains both acceptable and technically feasible for developing nations like Benin [25,27,43].

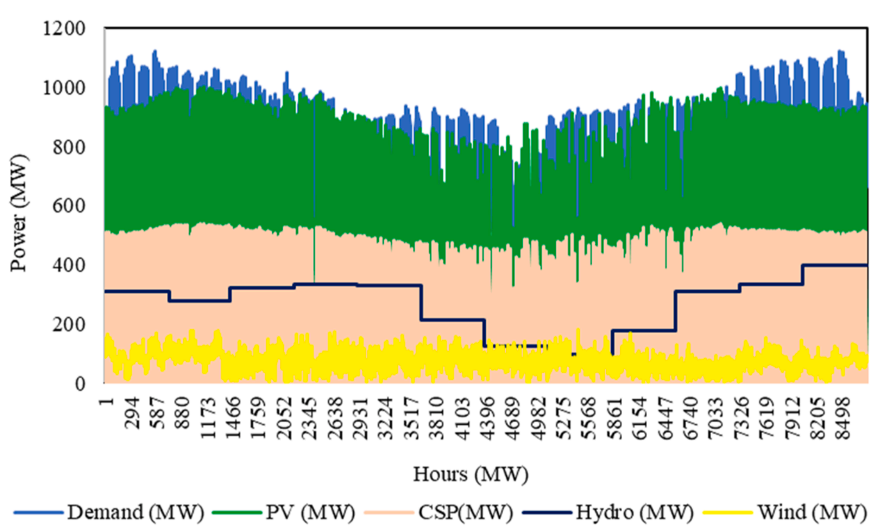

Fig. 15. Hourly electricity demand/supply for 100 % optimal RES integration using NG+PV+Wind +Hydro + CSP.

3.3. 50 % RE in 2050 scenario analysis

In this scenario, the assumed RE penetration level is set at 50 % by mid-century. Given Benin's current energy landscape, where RE accounts for just 3.2 % of the supply mix, 24.6 % by 2025 (in the coming two years), 44 % by 2030, and 100 % by mid-century appears overly ambitious. To reach a 50 % RE integration by 2050 would necessitate an investment cost of USD 4.3 billion, with an annual cost of USD 612 million. The envisioned supply mix involves the installation of 125 MW of solar PV, 600 MW of hydropower, 200 MW of wind energy, and 60 MW of CSP. This RE generation mix comprises 0.27 TWh from solar PV, 0.44 TWh from wind, 2.36 TWh from hydropower, and 0.14 TWh from CSP, with no CEEP generation, as detailed in Table 7. Comppared to government targets scenario and the 2 %RE per year scenario which generates 1242 and 276 MW of CEEP, this scenario is the best option that the government could consider. Nevertheless, a more realistic 50 % target remains achievable with the development of robust RE integration strategies and effective fund mobilization and management. Globally, various regions are committed to diversifying their electricity production sources to meet Sustainable Development Goal 7. For example, Nigeria aims for a 30 % RE integration by 2030 [41]. In the European Union (EU), the target is a minimum of 30 % RE in the electricity mix by 2030, while Denmark aspires to reach 50 % and 100 % RE by 2030 and 2050, respectively [27].

The study's findings demonstrate the role of RE integration in Benin's electricity supply mix. The falling cost of RE technologies is a means to accelerate the energy transition and decrease the impact of climate change [57]. The country's commitment to reducing its reliance on fossil fuels and addressing energy security concerns has driven these advancements. Solar CSP projects, particularly in the northern regions of the country, should gain traction, harnessing the intense sunlight to generate electricity. PV solar installations need also to be increased, with both grid-connected and off-grid systems helping to provide reliable power to rural and urban areas. The prospects for these renewable energy sectors in the Benin Republic are promising, as they not only address environmental concerns but also create opportunities for job growth and energy access for remote communities. Continued investment and policy support will be essential in realizing the full potential of these renewable technologies, making Benin a regional leader in sustainable energy production and consumption.

The total electricity demand of Nigeria is estimated to reach 200 TWh/yr in 2030, which is 46 times higher than that of Benin in the same year [41]. This implies that it will be easier for Benin to meet the 100 % electricity access target if the necessary investment strategies and policies are developed. In addition, Benin must join regional initiatives such as the West African Power Pool (WAPP), which promotes regional electricity access in West Africa to accelerate its electricity independence [34]. Developing strategies for integrating locally available RE sources is crucial to diversifying electricity sources while fostering energy transition.

Table 7
Scenario comparison for 2050.

Technology	Government target scenario (2050)	2 % RE growth per year scenario (2050)	50 % RE in 2050 scenario
NG (MW)	1124	414	563
PV (MW)	1000	410	125
Wind (MW)	300	300	200
Hydro (MW)	600	600	600
CSP (MW)	538	65	60
CEEP (MW)	1262	276	0
Investment cost (USD billion)	6.04	4.5	4.3

3.4. Analysis of electricity generation from MSW in urban cities in Benin

The MSW management is still challenging, and there is no electricity generation from the MSW in Benin. Nevertheless, a notable opportunity exists to harness valuable electricity from landfill biogas, offering the dual benefit of meeting energy demands while concurrently curbing CH_4 emissions. The results indicate that about 2.98 PJ, 4.14 PJ, and 14.83 PJ of gross energy could be produced from the different MSWs in Benin in 2025, 2030, and 2050, respectively (Table 8). This indicates that about 0.232 TWh/yr of electricity could be generated from MSW in 2025 in Benin. Moreover, this electricity production represents approximately 35 % of the 0.66 TWh/yr of RE electricity required to meet the 24.6 % RE in the supply mix.

Similarly, the outlook for 2030 and 2050 showcases substantial progress in electricity generation from MSW. By these years, the total electricity generation from MSW is expected to reach 0.3215 TWh and 1.16 Wh, constituting 19.6 % and 18.12 % of RE electricity necessary to achieve 44 % and 100 % RE electricity within Benin's energy supply mix, respectively. This underscores the substantial potential of waste-to-energy solutions for meeting energy demands and advancing Benin's renewable energy goals while concurrently addressing environmental concerns by mitigating harmful greenhouse gas emissions, particularly methane.

3.5. Discussion and policy implications

In this study, different scenarios are developed and analyzed to assist decision-makers take informed decisions for a sustainable electricity transition in Benin. The generation of electricity from NG and solar PV is the most economically viable option for the country, given its location and the abundant solar radiation received annually. Approximately 770 MW and 3002 MW of PV capacity would help meet 44 % and 100 % of electricity from RE by 2030 and 2050 respectively under the government target scenario. This would produce CEEP in the power system, which the development and promotion of electric vehicles would help to eliminate. As presented in Table 7, the government targets scenario seems too ambitious with the generation of 1242 MW of CEEP and require more financial investment. The two other scenarios correspond more with the current situation and space in the energy sector of Benin and are the most likely to happen options. Particularly, 50 % by 2050 requires less investment with no CEEP generation. It is noteworthy that it was only in 2022 that Benin inaugurated its first grid-tied solar PV plant [23]. In addition, the country's electricity supply mix mainly depends on electricity importation from neighboring countries such as Nigeria and Ghana [9], and local RE sources have not been exploited much. Increasing local electricity generation can reduce import dependency and transmission losses.

The feasibility of the presented scenarios in Benin's context requires new policy development and private-sector engagement. It is important to upgrade Benin's existing power grid to deploy large-scale solar PV and wind power systems. In addition, appropriate policy development, financial support, and intergovernmental collaboration are required to foster RE integration and ensure power system reliability. Investment in adequate infrastructure development could boost national electricity

Table 8
Gross energy production and electricity generation from MSW.

Category	Gross energy production E_{CH4} (PJ)			Electricity generation E_{el} (TWh/yr)		
MSW Category	2025	2030	2050	2025	2030	2050
Highly degradable	2.75	3.823	13.7	0.214	0.297	1.07
Moderately degradable	0.23	0.315	1.13	1.78×10^{-2}	2.45×10^{-2}	8.75×10^{-2}
Total	2.98	4.14	14.83	0.232	0.3215	1.16

production. Moreover, biomass-to-energy systems have not been investigated in Benin. For instance, the northern part of the country is endowed with agricultural and animal waste that can be harnessed for power generation [33]. The amount of waste collected annually by the Beninese company of waste collection and management (SDGS) could be converted into methane and used to produce electricity, as demonstrated by this study's results. This will contribute to increasing the share of RE in the country's electricity mix and reducing the methane released into the atmosphere yearly.

The government of Benin has initiated different projects to gradually integrate RE resources and ensure 100 % electricity access to Beninese by 2050 [23]. For instance, Benin's millennium challenge account (MCA) has launched several projects to promote the country's economic growth and electricity independence. in their nationally determined (NDC) report, the government has plans to build 128 MW, 60.2 MW, and 18.8 MW of hydropower plants at Dogo, Vossa, and Bétérou, respectively [15]. The projects also include off-grid and grid-connected solar PV systems and solar home systems for rural areas not connected to the national grid. These are crucial projects that, if effectively implemented, could increase the country's RE electricity production and help achieve the scenarios developed in this study.

According to Akpahou & Odoi-Yorke [5] CSP systems are promising in the Benin Republic, and further investigation is needed to boost the development of such technology in the country. Investing in research and development strategies would enhance the efficiency and affordability of CSP and wind technology. Adequate regulatory frameworks, including feed-in tariffs and tax incentives, can attract investors and motivate the population to invest in RE projects. Deploying RE sources will reduce reliance on fossil fuels and decrease carbon emissions, improving air quality and public health. The transition to RE sources such as solar PV, wind, and bioenergy would create job opportunities for the local population of Benin.

4. Conclusion

This study analyses the strategies for increasing RE electricity generation in Benin's energy supply by 2050. Three different scenarios were developed including the government targets scenario, 2 % RE integration per year scenario, and 50 % RE by 2050 scenario. In addition, electricity generation from MSW sources abundantly available in the country has been investigated. The main findings are as follows:

- Under the government targets scenario, NG could be used as a single energy source for electricity generation with an installation capacity of 474,657, and 1124 MW in 2025, 2030, and 2050 respectively, to meet demand and ensure 100 % electricity access. In addition, a combination of 370.4 MW of NG, 124 MW of PV, 87 MW of wind, 276 MW of hydropower, and 45 MW of CSP would provide 44 % of RE in the supply mix by 2030 under this scenario. To achieve 100 % RE by 2050, Benin would need to install 1000 MW of solar PV, 300 MW of wind power, 600 MW of hydropower, and 538 MW of CSP, with an investment cost of USD 6.04 billion and 1262 MW of CEEP generation.
- Moreover, under the 2 % per annual scenario, the country would need to install 414 MW of NG, 410 MW of PV, 300 MW of wind, 600 MW of hydro, and 65 MW of CSP plant. Such installation would require an investment cost and a total annual cost of USD 4.5 billion and USD 554 million respectively to achieve 63 % of RE by 2050. Furthermore, this scenario could generate 276 MW of CEEP with a lower investment cost compared to the government targets scenario. Increasing electricity generation from hydropower would eliminate the CEEP but require additional financial investment.
- Under the 50 % RE in 2050 scenario, USD 4.3 billion is required to install 563 MW of NG, 125 MW of PV, 200 MW of wind, 600 MW of hydropower, and 60 MW of CSP with no CEEP generation, to achieve 50 % RE by 2050. This scenario is the most economically feasible and likely to happen in the current situation of Benin if adequate RE integration plans are developed and implemented well.
- The total electricity generation from MSW in Benin is estimated to be 0.232, 0.3215, and 1.16 TWh/yr in 2025, 2030, and 2050, respectively. This would constitute approximately 35 %, 19.6 %, and 18.13 % of the RE electricity needed to meet the government targets for those years.

Hybridizing the energy sources will improve Benin's electricity supply and guarantee access to an adequate and reliable electricity supply. For the feasibility of the scenarios developed in this study, Benin needs to develop an adequate policy and regulatory framework for RE integration. Interdisciplinary collaboration and RE financing mechanisms would further accelerate the country's energy transition. Future research could investigate the transmission and distribution approaches required to realize the proposed scenarios.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Romain Akpahou: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Writing – original draft, Writing – review & editing. **Flavio Odoi-Yorke:** Conceptualization, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Lena D. Mensah:** Conceptualization, Project administration, Resources, Supervision, Validation, Writing – original draft. **David A. Quansah:** Conceptualization, Methodology, Resources, Supervision, Validation, Visualization, Writing – original draft. **Francis Kemausuor:** Conceptualization, Methodology, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

The authors acknowledge the Fostering Research & Intra-African Knowledge Transfer through Mobility & Education (FRAME) scholarship program funded by the European Union (project number: 2019–1973/007–001), for providing Mr. Romain Akpahou with a scholarship to pursue a Ph.D. program at Kwame Nkrumah University of Science and Technology (KNUST). The authors thank the two anonymous reviewers for their constructive feedback.

References

[1] IRENA, "Planning and prospects for renewable power: west Africa," 2018. [Online]. Available: https://www.irena.org/publications/2018/Nov/Planning-and-prospects-for-renewable-power.

[2] J.Z. Thellufsen et al., "Smart energy cities in a 100% renewable energy context," vol. 129, no. November 2019, 2020, http://doi.org/10.1016/j.rser.2020.109922.

[3] IRENA, "The renewable energy transition in africa powering access, resilience and prosperity," 2020, [Online]. Available: https://www.irena.org/publications/2021/March/The-Renewable-Energy-Transition-in-Africa (Accessed November 18, 2022).

[4] IEA, IRENA, UNSD, W. Bank, and WHO, Tracking SDG7: the energy progress report, World Bank, Washington DC, 2022, pp. 158–177. © World Bank. License: Creative Commons Attribution—NonCommercial 3.0 IGO (CC BY-NC 3.0 IGO).

[5] R. Akpahou, F. Odoi-yorke, A multicriteria decision-making approach for prioritizing renewable energy resources for sustainable electricity generation in Benin, Cogent Eng 10 (1) (2023), https://doi.org/10.1080/23311916.2023.2204553.

[6] Africa Energy Review, "Africa energy review 2021: the global race to net zero by 2050 is accelerating. Will Africa realise a just transition or become a stranded asset?," no. November, p. 37, 2021, [Online]. Available: www.pwc.co.za/energyreview.

[7] F. Odoi-yorke, R. Akpahou, R. Opoku, L.D. Mensah, Technical, financial, and emissions analyses of solar water heating systems for supplying sustainable energy for hotels in Ghana, Sol. Compass 7 (June) (2023) 100051, https://doi.org/10.1016/j.solcom.2023.100051.

[8] SEforALL, "Sustainable energy for all, Benin - SEforALL Africa Hub," 2022. https://www.se4all-africa.org/seforall-in-africa/country-data/benin/(Accessed November 24, 2022).

[9] J.H.R. Mensah, I.F.S. dos Santos, G.L. Tiago Filho, A critical analysis of the energy situation in the Benin Republic and its evolution over the last decade, Renew. Energy 202 (March 2022) (2022) 634–650, https://doi.org/10.1016/j.renene.2022.11.085.

[10] J.H.R. Mensah, et al., Assessment of electricity generation from biogas in Benin from energy and economic viability perspectives, Renew. Energy 163 (2021) 613–624, https://doi.org/10.1016/j.renene.2020.09.014.

[11] H.C. Sotiman Yotto, et al., The renewable energy landscape in Benin: an analysis and review of barriers, targets, policies and actions for a clean energy transition, in: Int. Conf. Electr. Comput. Commun. Mechatronics Eng. ICECCME 2021, 2021, pp. 7–8, https://doi.org/10.1109/ICECCME52200.2021.9590886.

[12] R. Akpahou, F. Odoi-yorke, L. Kwasi, Techno-economic analysis of a utility-scale grid-tied solar photovoltaic system in Benin republic, Clean. Eng. Technol. 13 (April) (2023) 100633, https://doi.org/10.1016/j.clet.2023.100633.

[13] MERPMDER, "Plan d'actions national des energies renouvelables du Bénin (PANER - BENIN)," 2015. [Online]. Available: https://www.se4all-africa.org/fileadmin/uploads/se4all/Documents/Country_PANER/Benin_Plan_d_Actions_National_pour_les_Energies_Renouvelables__PANER_.pdf.

[14] O.D.T. Odou, R. Bhandari, R. Adamou, Hybrid off-grid renewable power system for sustainable rural electrification in Benin, Renew. Energy 145 (2020) 1266–1279, https://doi.org/10.1016/j.renene.2019.06.032.

[15] MCVDD, "Nationally determined contribution of Benin under the Paris agreement. updated version (Final document). (French)," 2021.

[16] D. Bignon et al., "Socio-demographic and economic characteristics, crop-livestock production systems and issues for rearing improvement : a review," vol. 12, no. February, pp. 519–541, 2018.

[17] I. Pappis et al., "Energy projections for African countries - publications office of the EU," 2019. http://doi.org/10.2760/678700.

[18] IRENA, "Country Indicators and Sdgs: benin.," *Masdar City, Abu Dhabi*, 2022. https://www.irena.org/-/media/Files/IRENA/Agency/Statistics/Statistical_Profiles/Africa/Benin_Africa_RE_SP.pdf (Accessed November 30, 2022).

[19] O.O. Ajayi, Sustainable energy development and environmental protection: implication for selected states in West Africa, Renew. Sustain. Energy Rev. 26 (2013) 532–539, https://doi.org/10.1016/j.rser.2013.06.009.

[20] SDGS, "Management of household solid waste," 2022. https://sgds.bj/nos-activites/gestion-des-dechets/ (accessed Sep. 23, 2023).

[21] SIE-Benin, "Key figures 2021, energy balances and indicators 2016 to 2020. (French).," pp. 1–26, 2021, [Online]. Available: https://direction-energie.gouv.bj/actualites/sie-benin-atelier-national-de-presentation-du-systeme-dinformation-energetique-du-benin-3.html.

[22] A.B. Yatokpa, "Benin national action plan for energy efficiency panee. (French).," 2015. [Online]. Available: http://documents.worldbank.org/curated/en/2015/07/24819836/eeea-enhancing-world-bank?s-energy-access-investments-through-energy-efficiency.

[23] R. Akpahou, L.D. Mensah, D.A. Quansah, Renewable energy in Benin : current situation and future prospects, Clean Energy 7 (5) (2023) 952–961.

[24] A.S. Oyewo, A. Aghahosseini, M. Ram, A. Lohrmann, C. Breyer, Pathway towards achieving 100 % renewable electricity by 2050 for South Africa, Sol. Energy 191 (October 2018) (2019) 549–565, https://doi.org/10.1016/j.solener.2019.09.039.

[25] D. Connolly, H. Lund, B.V. Mathiesen, M. Leahy, The first step towards a 100 % renewable energy-system for Ireland, Appl. Energy 88 (2) (2011) 502–507, https://doi.org/10.1016/j.apenergy.2010.03.006.

[26] D. Groppi, B. Nastasi, M.G. Prina, D.Astiaso Garcia, The EPLANopt model for Favignana island's energy transition, Energy Convers. Manag. 241 (2021) 114295, https://doi.org/10.1016/j.enconman.2021.114295.

[27] H. Lund, B.V. Mathiesen, Energy system analysis of 100 % renewable energy systems-The case of Denmark in years 2030 and 2050, Energy 34 (5) (2009) 524–531, https://doi.org/10.1016/j.energy.2008.04.003.

[28] B. Ćosić, G. Krajačić, N. Duić, A 100 % renewable energy system in the year 2050: the case of Macedonia, Energy 48 (1) (2012) 80–87, https://doi.org/10.1016/j.energy.2012.06.078.

[29] J. Graça Gomes, J. Medeiros Pinto, H. Xu, C. Zhao, H. Hashim, Modeling and planning of the electricity energy system with a high share of renewable supply for Portugal, Energy 211 (2020), https://doi.org/10.1016/j.energy.2020.118713.

[30] D. Icaza-Alvarez, P. Arias Reyes, F. Jurado, M. Tostado-Véliz, Smart strategies for the penetration of 100 % renewable energy for the Ecuadorian Amazon region by 2050, J. Clean. Prod. 382 (August 2022) (2023), https://doi.org/10.1016/j.jclepro.2022.135298.

[31] M. Izanloo, Y. Noorollahi, A. Aslani, Future energy planning to maximize renewable energy share for the south Caspian Sea climate, Renew. Energy 175 (2021) 660–675, https://doi.org/10.1016/j.renene.2021.05.008.

[32] S.V... Dedjinou, Electricity demand from households in Benin (La Demande D'Énergie Électrique Des Ménages Au Benin), Les. Cah. Du. Cread. 38 (1) (2022) 61–104, https://doi.org/10.4314/cread.v38i1.3.

[33] D.G.F. Adamon, A.A. Jossou, A.N. Adomou, F. Pinta, Evaluation of the energy potential of agricultural waste in West Africa from three biomasses of interest in Benin, Int. J. Adv. Res. 8 (4) (2020) 766–774, https://doi.org/10.21474/ijar01/10839.

[34] H.P. Ishaku, H. Adun, M. Jazayeri, M. Kusaf, Decarbonisation strategy for renewable energy integration for electrification of West African Nations: a bottom-up energyplan modelling of west African power pool targets, Sustain 14 (23) (2022), https://doi.org/10.3390/su142315933.

[35] P.A. Kwakwa, Analysis of the determinants of electricity consumption in Benin, Res. Artic. J. Energy Manag. Technol. 2 (3) (2018) 42, https://doi.org/10.22109/jemt.2018.125839.1075 [Online]. Available:.

[36] F.J.-L. Comlan, S.G. Clarence, D. Emmanuel, A.D. K, G. Gérard, Simulation of the injection of 25 MW photovoltaic energy production: analysis of the impacts on the grid of the société béninoise d'energie electrique (SBEE), IRA-International J. Appl. Sci. (ISSN 2455-4499) 16 (2) (2021) 16, https://doi.org/10.21013/jas.v16.n2.p1.

[37] D.W. Atchike, Z.Y. Zhao, G. Bao, The relationship between electricity consumption, foreign direct investment and economic growth: case of Benin, Int. J. Energy Econ. Policy 10 (4) (2020) 507–515, https://doi.org/10.32479/ijeep.9415.

[38] D.T.A. Marcel, Electricity consumption and economic growth nexus in the republic of Benin, Socioecon. Challenges 3 (2) (2019) 63–69, https://doi.org/10.21272/sec.3(2).63-69.2019.

[39] Global Solar Atlas, "Benin solar resource map," 2023. https://globalsolaratlas.info/download/benin (accessed Dec. 10, 2023).

[40] E. Mboumboue, D. Njomo, Biomass resources assessment and bioenergy generation for a clean and sustainable development in Cameroon, Biomass Bioenergy 118 (2018) 16–23, https://doi.org/10.1016/j.biombioe.2018.08.002. Nov.

[41] O. Bamisile, et al., An approach for sustainable energy planning towards 100% electrification of Nigeria by 2030, Energy 197 (2020) 117172, https://doi.org/10.1016/j.energy.2020.117172.

[42] H. Lund, J.Z. Thellufsen, and H. Lund, "EnergyPLAN – advanced energy systems analysis computer model (Version 16.2).," 2022. [Online]. Available: https://doi.org/10.5281/zenodo.6602938.

[43] B. Vad, H. Lund, K. Karlsson, 100% Renewable energy systems, climate mitigation and economic growth, Appl. Energy 88 (2011) (2015) 488–501, https://doi.org/10.1016/j.apenergy.2010.03.001.

[44] M. EL-Shimy, Optimal site matching of wind turbine generator: case study of the Gulf of Suez region in Egypt, Renew. Energy 35 (8) (2010) 1870–1878, https://doi.org/10.1016/j.renene.2009.12.013.

[45] I.T. Oloya, T.J.L. Gutu, M.S. Adaramola, Techno-economic assessment of 10 MW centralised grid-tied solar photovoltaic system in Uganda, Case Stud. Therm. Eng. 25 (December 2020) (2021), https://doi.org/10.1016/j.csite.2021.100928.

[46] EnergyPLAN, "EnergyPLAN Cost Database," 2018. https://www.energyplan.eu/useful_resources/costdatabase/ (accessed Aug. 24, 2023).

[47] The Worldbank, What-a-waste-report, in: World Bank Urban Dev. Ser. Knowl. Pap., 2012, pp. 1–116 [Online]. Available: https://openknowledge.worldbank.org/handle/10986/17388.

[48] World Bank, "Handbook for the preparation of Landfill gas to energy projects in Latin America and the Caribbean," 2004. [Online]. Available: https://www.esmap.org/sites/esmap.org/files/Handbook_Preparation_LandfillGas_to_EnergyProjects_LAC_Resized.pdf.

[49] I.F. Silva dos Santos, N.D. Braz Vieira, L.G.B. de Nóbrega, R.M. Barros, G.L. Tiago Filho, Assessment of potential biogas production from multiple organic wastes in Brazil: impact on energy generation, use, and emissions abatement, Resour. Conserv. Recycl. 131 (November 2017) (2018) 54–63, https://doi.org/10.1016/j.resconrec.2017.12.012.

[50] R. Arthur, M.F. Baidoo, G. Osei, L. Boamah, S. Kwofie, Evaluation of potential feedstocks for sustainable biogas production in Ghana: quantification, energy generation, and CO2 abatement, Cogent Environ. Sci. 6 (1) (2020), https://doi.org/10.1080/23311843.2020.1868162.

[51] N. Scarlat, V. Motola, J.F. Dallemand, F. Monforti-Ferrario, L. Mofor, Evaluation of energy potential of municipal solid waste from African urban areas, Renew. Sustain. Energy Rev. 50 (2015) 1269–1286, https://doi.org/10.1016/j.rser.2015.05.067.

[52] INStaD, "Population projections from 2014 to 2063 and outlook for social demand from 2014 to 2030 in Benin. (French)," 2022.

[53] IEA, "Net zero by 2050 a roadmap for the global energy sector," 2021. [Online]. Available: www.iea.org/t&c/noneedfor.

[54] GoG, "Government of the Gambia ministry of environment, climate change and natural resources," 2022. [Online]. Available: https://gisgeography.com/gambia-map/.

[55] SeeforAll, "Ghana energy transition and investment plan | sustainable energy for all," 2023. https://www.seforall.org/our-work/initiatives-projects/energy-transition-plans/ghana (accessed Oct. 09, 2023).

[56] S. Teske, S. Niklas, and S. Miyake, "Technical scenario for 100 % renewable energy in Nepal by 2050: possible transition pathways for NDC & LTS implementation; prepared for world future council, WWF and brot für die welt by the university of technology Sydney, institute for sustainable future," no. February 2023, [Online]. Available: www.isf.uts.edu.au.

[57] International Renewable Energy Agency, "Renewable generation costs in 2022. Abu Dhabi.," 2022.